

MAYANK SINGH

mayanksingh5399@gmail.com | +91 9651032034 | LinkedIn | GitHub

Summary

Full Stack Engineer with 3+ years experience in backend engineering, microservices, REST APIs, and scalable full stack application development. Strong foundation with experience building AI-powered enterprise solutions using LLMs, semantic search, vector embeddings, workflow automation and cloud-native architectures using AWS.

Skills

Languages: JavaScript, Java, MongoDB, C++, SQL
Dev: React, Node.js, Express, Redux, REST APIs, Microservices, HTML5, CSS3
AI/ML: RAG, Semantic Search, Vector Embeddings, LLMs, LangChain, LangGraph
Tools: AWS (Lambda, DynamoDB, API Gateway, S3, BedRock), Git, SonarQube, Postman, Vector Databases

Work Experience

AI Engineer

NimbleWork

Jan 2025 – Present

Bengaluru

- Integrated GPT-4o powered intelligent content generation capabilities into enterprise project management workflows, enabling **automated generation** of structured field data and improving user productivity across multiple project modules.
- Developed **AI-driven semantic analysis pipelines** to analyze user comments, feedback, and activity data across different card types, enabling contextual insights, **sentiment** understanding, and improved **decision-making workflows**.
- Automated development workflows using **Claude Code** integrations, accelerating engineering productivity and improving adherence to **TDD practices** across feature development cycles.
- Implemented AI-assisted static code analysis and automated **SonarQube** issue resolution workflows prior to **CI/CD** execution, significantly reducing code quality violations before pipeline validation.

Software Engineer

NimbleWork

Jan 2023 – Dec 2024

Bengaluru

- Delivered customer-facing features for a work management product using **React.js, Redux, and Java**, improving user experience and increasing daily active users by 30% across a multi-tenant SaaS platform serving 5K+ users.
- Architected a **microfrontend** module integrated with Java backend services, enabling independent **frontend** deployment and reducing latency by an estimated 25%.
- Designed and shipped **RESTful APIs** across **microservices** using Java, enabling seamless data flow between services and reducing average API response time by 20%.

Projects

VideoStreaming Platform – Semantic Search & Recommendations

Demo | Node.js, Express, Vector Search, AWS Bedrock, S3

- Engineered an AI-powered **semantic video search system** using **Amazon Bedrock** and vector embeddings, improving search relevance by 40% and enabling context-aware content discovery at scale.
- Solved poor content discoverability by enriching **embeddings with metadata**, increasing semantic match accuracy by 30% and user engagement across search queries using **vector search pipeline**.
- Developed scalable backend APIs using Node.js and Express to handle video uploads, semantic search, likes/dislikes, subscriptions, supporting **real-time indexing** and **retrieval of content**.

GTBuddy – Restaurant Management SaaS

Demo | React.js, Node.js, AWS Lambda, DynamoDB, WebSockets

- Designed and developed **billing, order management, and inventory** modules using **React.js and Node.js**, supporting restaurants, cafes, cloud kitchens, and food trucks across multiple outlets.
- Built a real-time **Kitchen Order Ticket (KOT)** and **table management** system using WebSockets, reducing kitchen order errors by an estimated 35%.
- Deployed scalable backend infrastructure on **AWS**, reducing manual operational overhead by an estimated 40% and enabling multi-outlet management from a single **dashboard**.

Education

Indian Institute of Information Technology Guwahati

B.Tech in Computer Science | GPA: 8.44 / 10.0

Aug 2019 – June 2023

Coursework: Data Structures, Algorithms, Computer Networks, Database Management, Systems Programming